Falcon 9 First Stage Landing Prediction

Data-Driven Approach to Predicting Rocket Reusability

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EXECUTIVE SUMMARY

Can we predict whether a Falcon 9 booster will land successfully and price a rival bid accordingly?

- SpaceX lists Falcon 9 launches at **$62M** vs. competitors at **$165M+**. The cost advantage comes almost entirely from first-stage reusability. Hence, predicting landing success = predicting true launch cost.
- **Data:** 90 Falcon 9 launch records collected via SpaceX REST API and Wikipedia web scraping (2010–2021).
- **Methodology:** Data collection → Wrangling → EDA (visualization + SQL) → Interactive analytics (Folium + Plotly Dash) → Machine learning classification.
- **Key Finding:** KSC LC-39A has the highest success rate among all sites. Higher flight numbers correlate strongly with successful landings, indicating an experience curve effect.

INTRODUCTION

- SpaceX disrupted the launch industry by pioneering reusable rockets.
- The Falcon 9's first stage, if successfully recovered, can be refurbished and re-flown — dramatically reducing per-launch costs.
- If a competing aerospace company wanted to bid against SpaceX, accurately estimating launch costs requires knowing the probability of a successful first-stage landing.
- Problems we set out to answer:
  - Which factors most influence landing success?
  - Are there geographic or operational patterns that explain success/failure?
  - Can we build a machine learning model that reliably predicts whether a booster will land successfully?

METHODOLOGY

Data collection
- SpaceX API
- Wikipedia

Data Wrangling
- Cleaning data
- Handling Null values

EDA
- Visualization
- SQL analytics

Interactive Analysis
- Folium Maps
- Plotly Dashboard

Predictive Modeling
- Logistic Regression, SVM
- Decision Tree, KNN

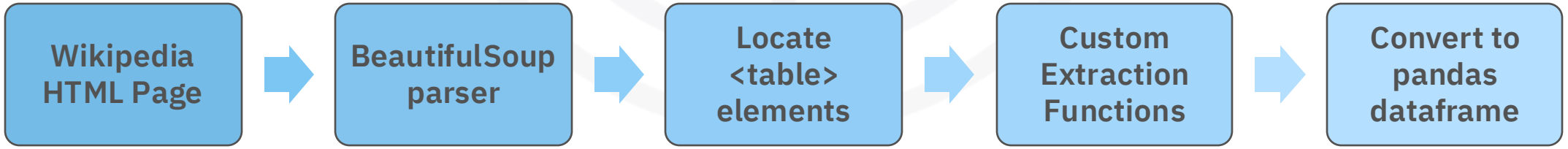
Data Overview

- https://github.com/binay-99/spacex-falcon9-landing-prediction/blob/main/notebooks/data_collection_api.ipynb
- Two complementary data sources were used to build a comprehensive 90-record dataset
- SpaceX REST API provided structured launch metadata (booster version, payload, orbit, outcome)
- Wikipedia web scraping supplemented with historical records not available via API
- Data spans **2010–2021** covering the complete history of Falcon 9 operations
- GET /v4/launches (Falcon 9 filter)

```
│
├─── GET /rockets/{id}      → BoosterVersion
├─── GET /launchpads/{id}   → LaunchSite, Lat, Long
├─── GET /payloads/{id}     → PayloadMass, Orbit
└─── GET /cores/{id}       → LandingOutcome, LandingPad, ReusedCount
        │
 [dataset_part_1.csv — 90 rows × 17 columns]
```

Data Overview

- https://github.com/binay-99/spacex-falcon9-landing-prediction/blob/main/notebooks/webscraping.ipynb
- Loaded raw dataset (90 launches, 17 features); identified columns with missing values
- `LandingPad` column has **26** missing values — acceptable since not all missions attempted a pad landing (ocean landings or no landing attempt)
- All other 16 columns were worked to having **0%** missing values
- Dataset has mix of numerical and boolean/categorical columns
- The `Outcome` column contained 8 distinct outcomes. These were consolidated into a binary column **Class** with value 1 for successful landing and 0 for failure or no attempt
- Data wrangling notebook:https://github.com/binay-99/spacex-falcon9-landing-prediction/blob/main/notebooks/Data_wrangling.ipynb



RESULTS

- EDA revealed a clear experience curve: success rates climbed from near 0% in 2012 to ~100% by 2020, driven by booster iteration and operational learning
- Interactive analytics confirmed KSC LC-39A as the highest-performing site, with coastal proximity and infrastructure as deliberate design choices
- Three classification models (LR, SVM, KNN) converged at 83.3% test accuracy — a robust, generalizable prediction ceiling for the current feature set

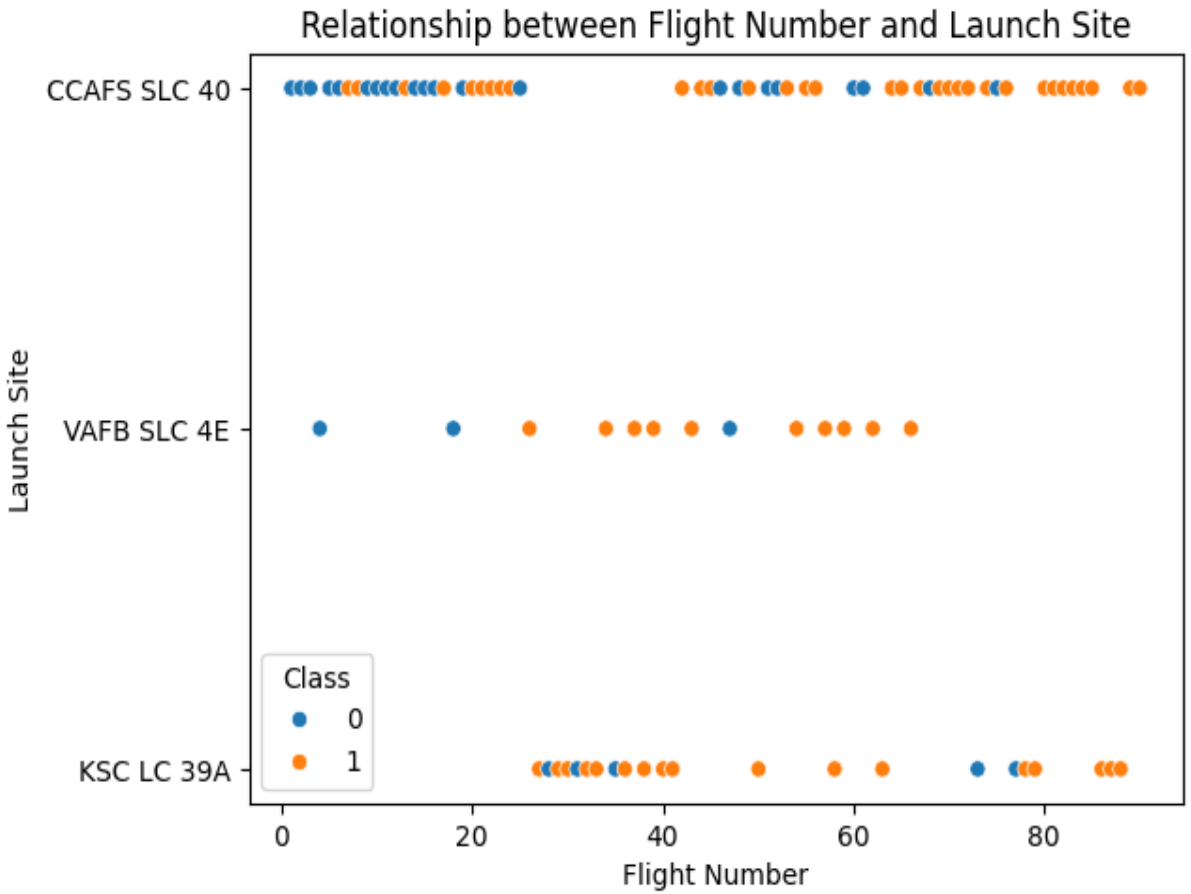
EDA: Data Visualization

- https://github.com/binay-99/spacex-falcon9-landing-prediction/blob/main/notebooks/eda_data_viz.ipynb

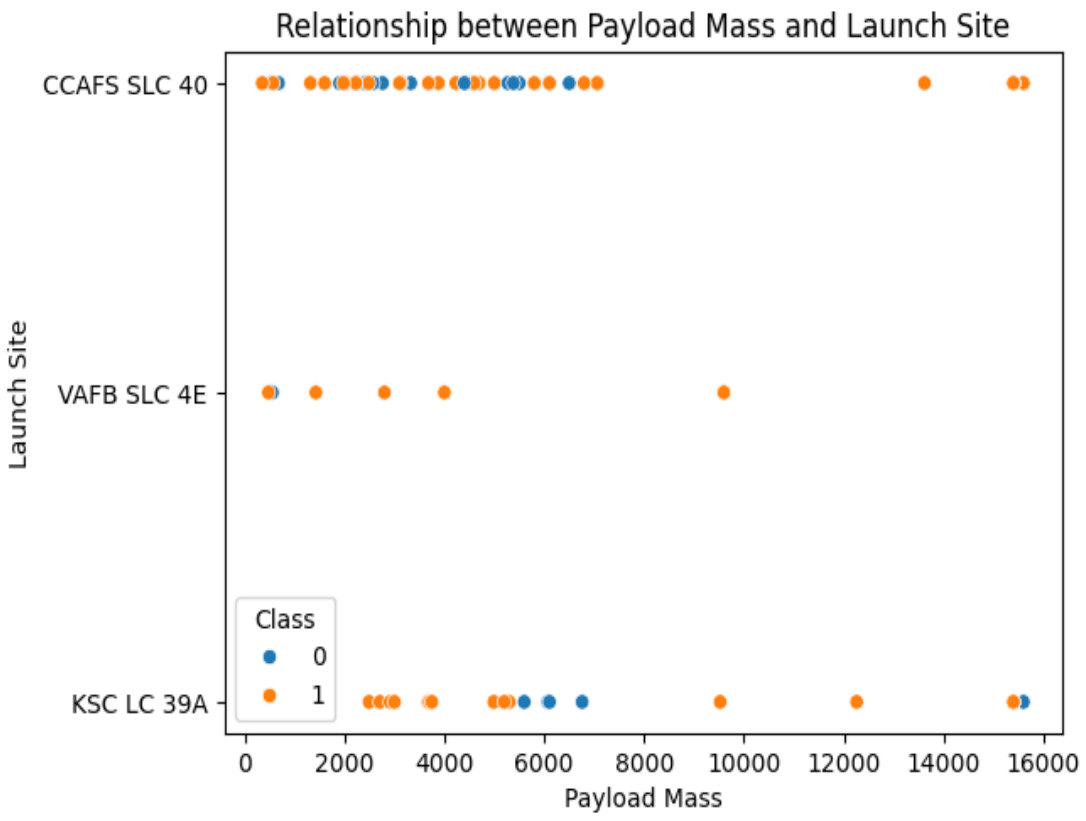
Chart	Purpose
Scatter: Flight Number vs. Launch Site	Reveal if later flights (more experience) succeed more
Scatter: Payload Mass vs. Launch Site	Identify payload thresholds per site
Bar: Success Rate by Orbit Type	Find which orbits are most predictable
Scatter: Flight Number vs. Orbit Type	Check if orbit-specific experience accumulates
Scatter: Payload Mass vs. Orbit Type	Find payload-orbit combinations that fail
Line: Yearly Average Success Rate	Show the learning curve over time

EDA: Data Visualization

- Early flights (Flight Number <20) are predominantly from CCAFS SLC-40, with mixed success.
- As flight numbers increase past 60+, KSC LC-39A dominates and shows mostly successful outcomes.
- VAFB SLC-4E shows fewer launches but reasonable success at mid-range flight numbers.



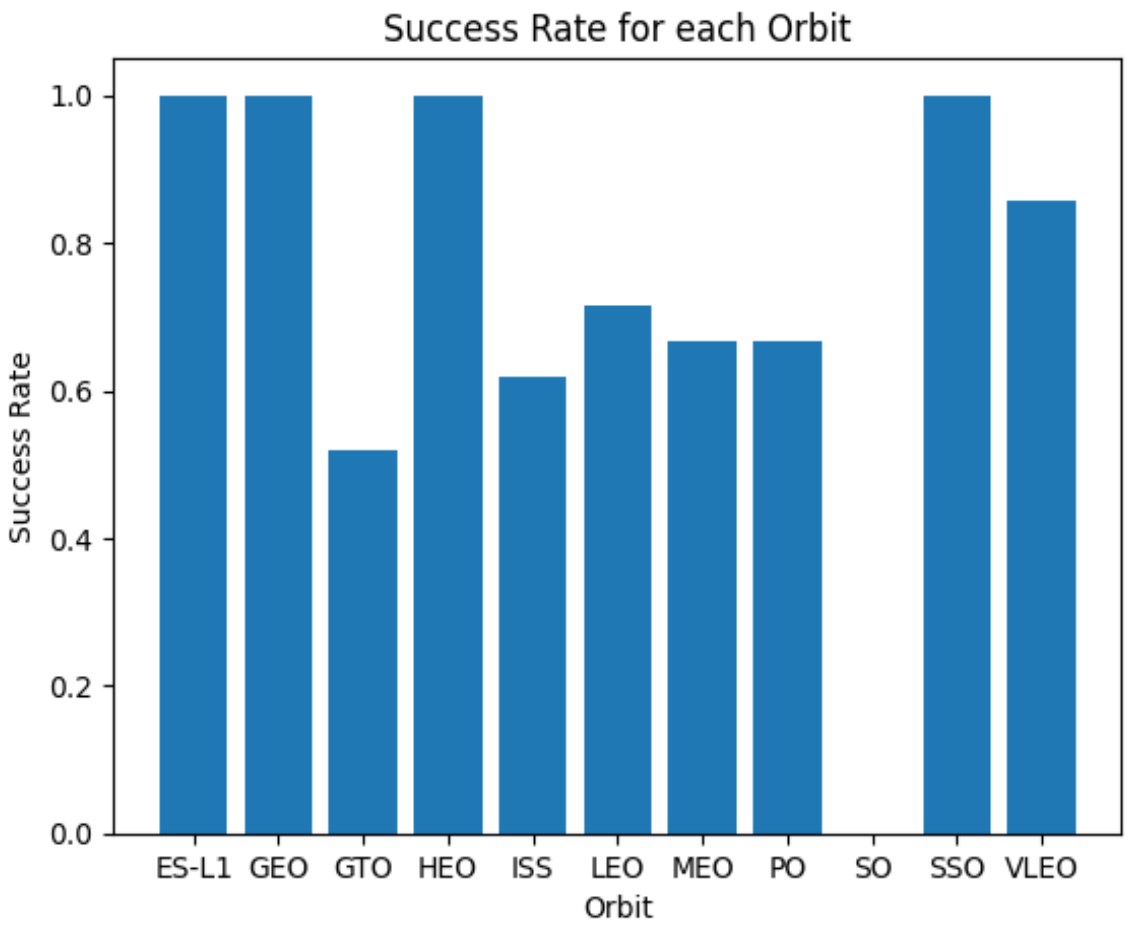
EDA: Data Visualization



- CCAFS SLC-40 handles the full payload range while KSC LC-39A handles heavier payloads
- VAFB SLC-4E is used for lighter payloads and polar orbits
- No site shows consistent failures at any particular payload range alone — suggesting payload interacts with orbit type.

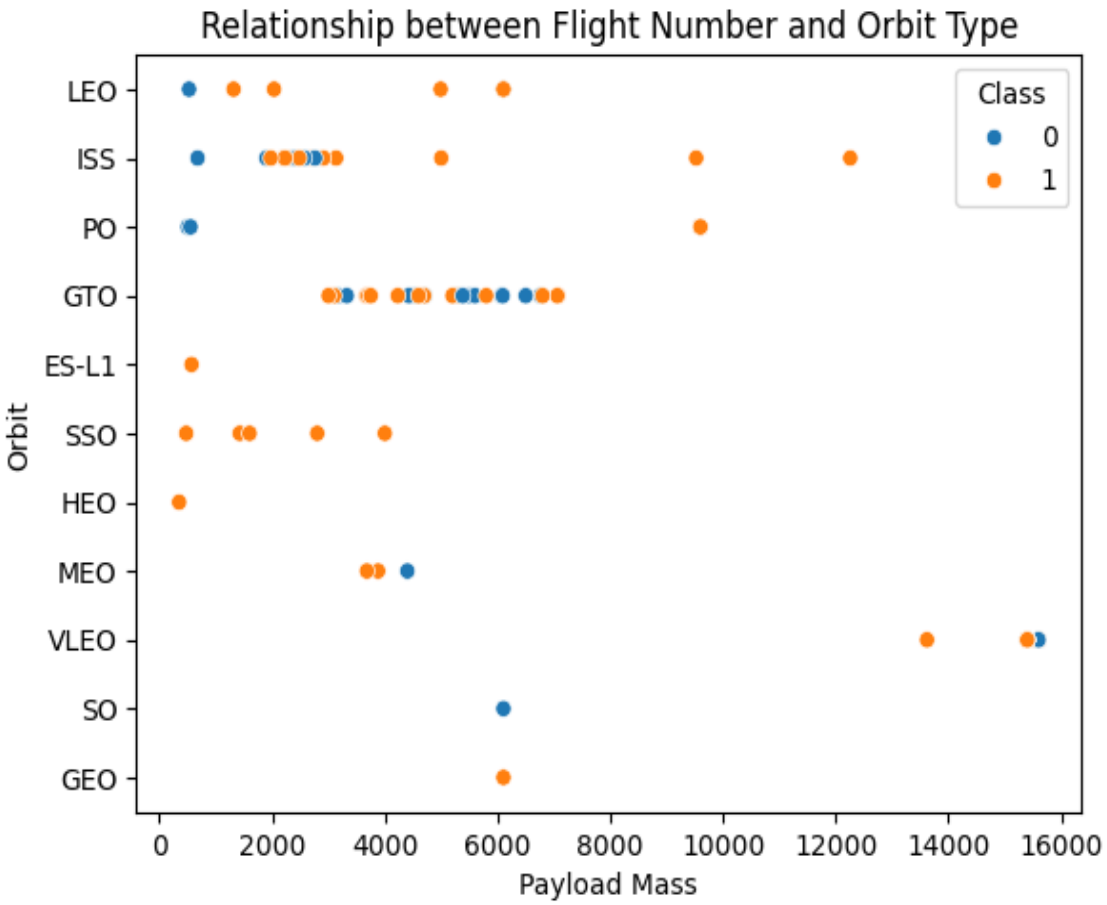
EDA: Data Visualization

- ES-L1, GEO, HEO, and SSO orbits show **100%** success rates — but with very few launches, so treat with caution.
- ISS and VLEO orbits have high but not perfect success rates.
- GTO has the lowest success rate, suggesting higher-energy missions were riskier in early years.



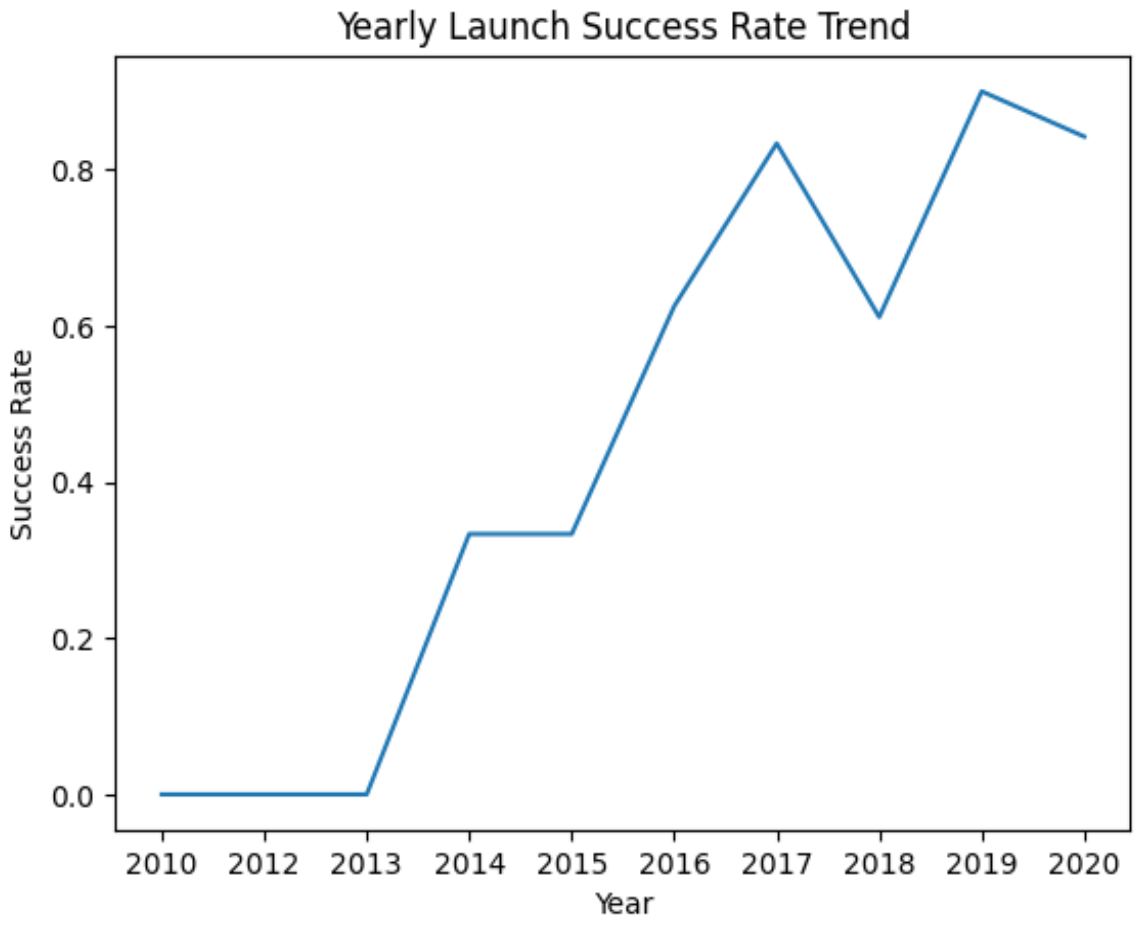
EDA: Data Visualization

- VLEO missions carry the widest payload range.
- GTO missions (often heaviest payloads, 5,000+ kg) show more failures, suggesting the combination of high payload + high-energy orbit stressed early booster recovery.
- Later GTO missions succeed as booster improvements were made.



EDA: Data Visualization

- Success rate was ~0% in 2010–2012 (early experimental phase). Climbed significantly from 2013–2016.
- By 2019–2020, success rate approached **100%** — demonstrating the compounding effect of engineering iteration and operational experience.



EDA: SQL Findings

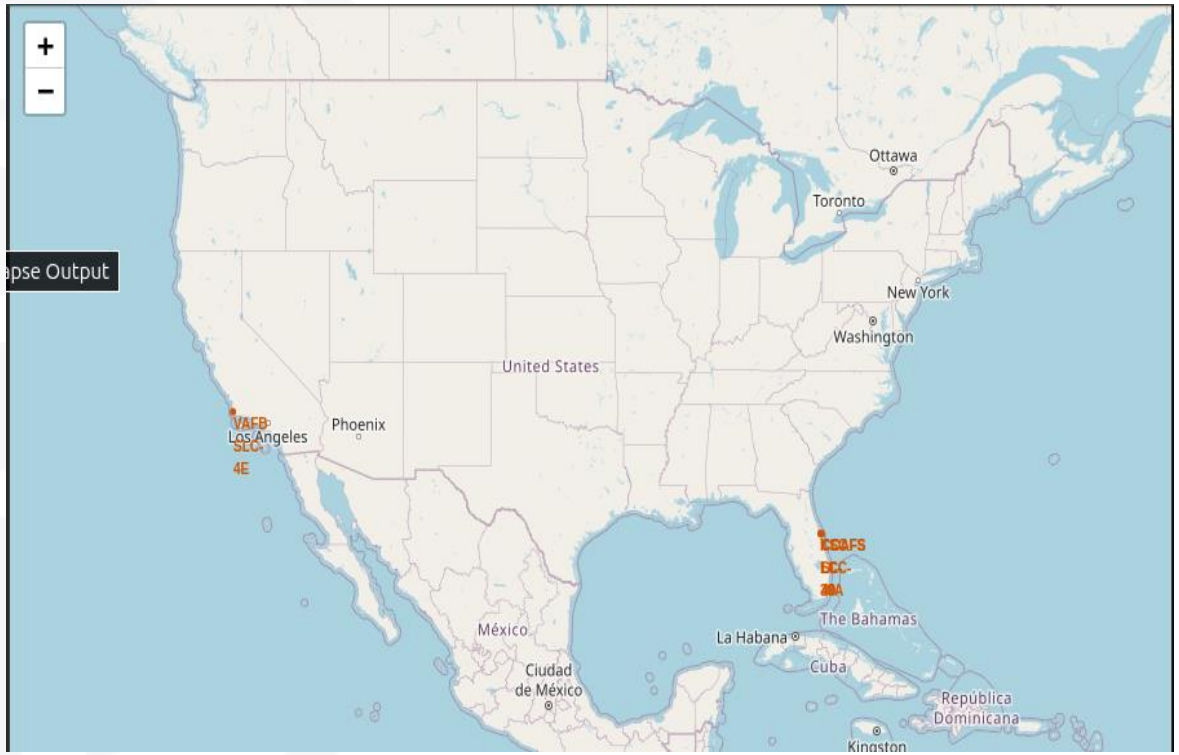
- SpaceX operates from two Florida complexes (Cape Canaveral) and one California site, covering both equatorial and polar orbit capability.
- NASA's Commercial Resupply Services contract was SpaceX's primary early customer. The total payload delivered to ISS demonstrates the mission-critical nature of these flights.
- F9 v1.1 was the workhorse mid-generation booster. Its average payload : ~2535 kg, gives a baseline for comparing against upgraded Block 5 boosters.
- Drone ship (ASDS) landings are used for heavier payloads going to high-energy orbits where the booster can't fly back.

EDA: SQL Findings

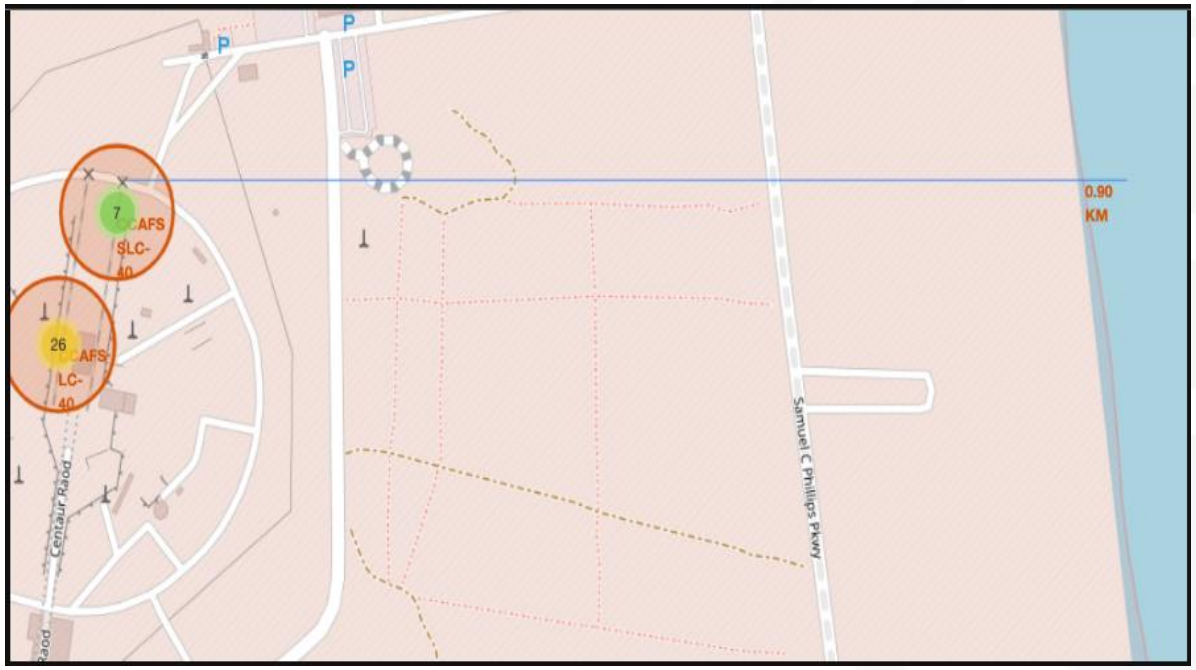
- Typically Block 5 boosters, SpaceX's most capable and most reused variant, handled the heaviest payloads in the dataset.
- 2015 was a critical learning year for drone ship landings. Analysis reveals the early challenges — SpaceX was still mastering the grid fin control and engine relight sequences needed for precision ASDS landings.
- The SQL analysis notebook can be found at: https://github.com/binay-99/spacex-falcon9-landing-prediction/blob/main/notebooks/sqllite.ipynb

Launch Site Locations

- https://github.com/binay-99/spacex-falcon9-landing-prediction/blob/main/notebooks/launch_site_location.ipynb
- All launch sites are situated on coastlines — essential for safety (rockets fly over ocean) and for RTLS landing on nearby pads
- Florida sites (CCAFS/KSC) enable low-inclination equatorial orbits to ISS and GEO
- VAFB in California enables polar and sun-synchronous orbits



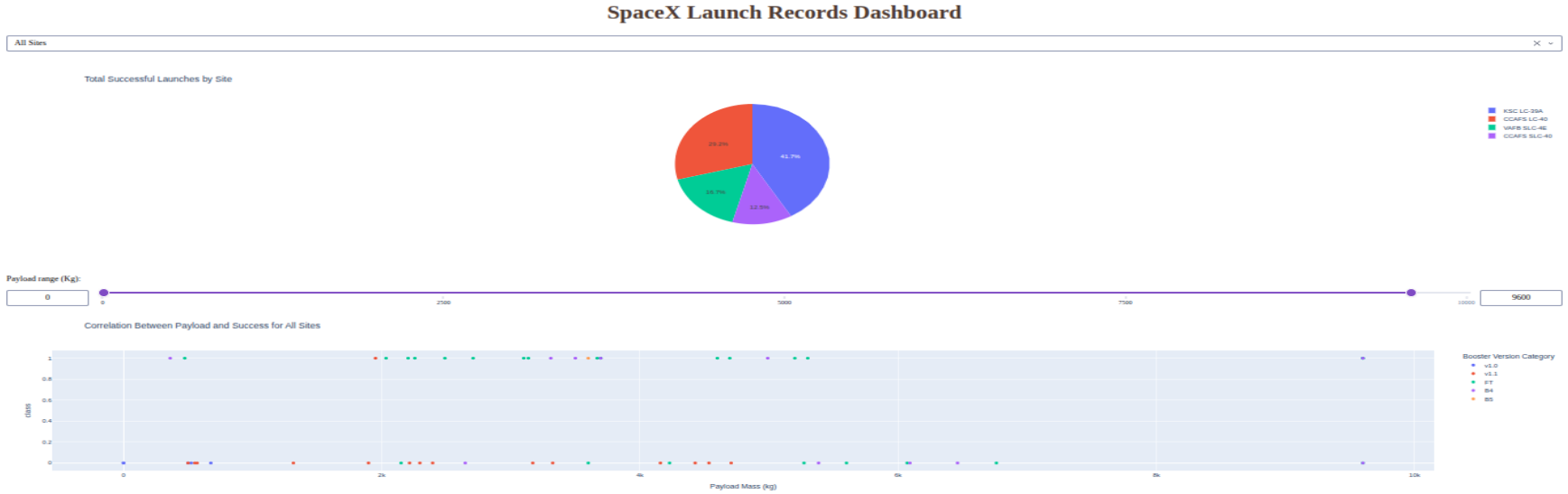
Launch Site Proximity Analysis



- KSC LC-39A is approximately **0.5 km** from the coastline — enables safe ocean-direction launches
- Proximity to highway infrastructure (SR-528) enables heavy payload transport
- Sites deliberately maintain distance from populated areas while remaining accessible for logistics

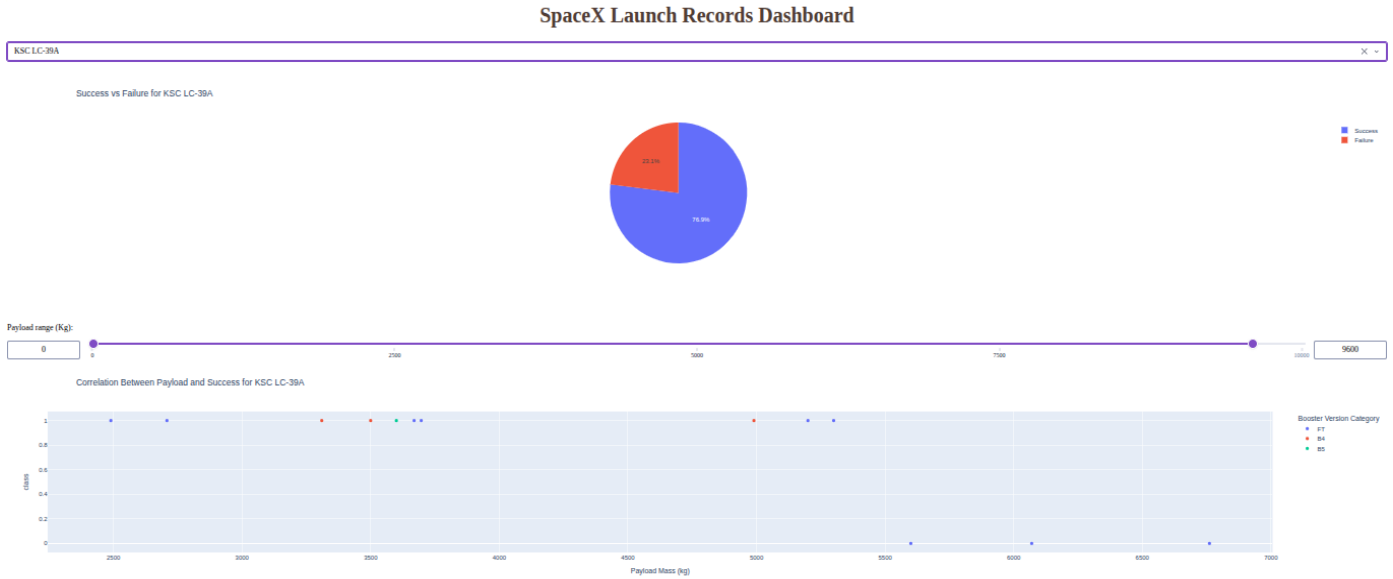
Total Launch Success by Site

- CCAFS SLC-40 has the most total launches due to being SpaceX's primary site since 2010.
- KSC LC-39A, though fewer total launches, has the highest proportion of successes. VAFB SLC-4E contributes the fewest launches.



Best-Performing Site Success Ratio

- KSC LC-39A shows the highest success-to-failure ratio among all sites.
- https://github.com/binay-99/spacex-falcon9-landing-prediction/blob/main/dashboard/spacex_dash_app.py



Classification Model Comparison

- Three models (LR, SVM, KNN) all converge at 83.3% test accuracy with strong generalization.
- The Decision Tree achieved the highest cross-validation score (87.7%) but shows signs of overfitting with a drop to 61.1% on unseen test data.
- For deployment, Logistic Regression or SVM are preferred for their stability.

Model	CV accuracy	Test accuracy
Logistic Regression	84.6%	83.3%
SVM	84.8%	83.3%
KNN	84.8%	83.3%
Decision Tree	87.7%	61.1%

CONCLUSION

- SpaceX's success rate is primarily a function of experience — flight number is the strongest single predictor of landing outcome
- KSC LC-39A is the optimal launch site: newest infrastructure, heaviest payloads, highest success ratio
- Payload mass alone doesn't predict failure — it's the combination of payload, orbit type, and booster generation that matters.

CONCLUSION

- At 83.3% test accuracy, the model is business-viable: a competing bidder can reliably estimate whether SpaceX will recover its booster and price accordingly
- The Decision Tree's overfitting (87.7% CV → 61.1% test) is itself a finding — it confirms the dataset is small enough that regularized models outperform complex ones

APPENDIX

- Feature list after one-hot encoding: FlightNumber, PayloadMass, Flights, GridFins, Reused, Legs, Block, ReusedCount + encoded Orbit, LaunchSite, LandingPad, Serial columns
- Model hyperparameters:
  - LR: C=0.01, penalty=l2, solver=lbfgs
  - SVM: C=1.0, gamma=0.0316, kernel=sigmoid
  - Decision Tree: criterion=entropy,max_depth=12, max_features=sqrt, min_samples_leaf=2, min_samples_split=5
  - KNN: algorithm=auto, n_neighbors=10, p=1(Manhattan distance)